AI-assisted comparison note

The common web and its individual coefficients

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Overview

We used an annular web to disprove Guralnick's conjecture on first cohomology for finite simple groups. Bowman, de la Fuente, Libedinsky and Plaza used a product of parabolic symmetrizers to prove unboundedness in every fixed positive cohomological degree. Our two groups arrived at these constructions independently. Here we compare them in degree one, where both lead to ordinary affine Kazhdan–Lusztig coefficients with the same lower endpoint.

In type A, singular Bott–Samelson characters and the corresponding split and merge webs describe the same normalized maps in the affine Schur category. Appending two normalized parabolic symmetrizers to their uncorrected product $\mathcal{BS}_m$ gives our six-stage web $\mathcal{R}_m$, in every rank and every standard coordinate. Each finite part is a single canonical element, and both constructions apply the same affine prefix.

Our web calculation gives the row sum $\sum_{i=1}^{m} \mu_{i;m} = \binom{m+1}{2}$. Their parabolic calculation gives the diagonal entry $\mu_{m;m} = m$ and follows the leading coefficient through higher degrees. To recover the individual entries, we complete the first j canonical constituents of their uncorrected product at fixed ambient rank. We obtain

$$\sum_{i=1}^{j} \mu_{i;m} = \binom{j+1}{2}, \qquad \mu_{j;m} = j \quad (1 \leq j \leq m).$$

For the partial sums we use their parabolic orbit estimates, our support, multiplicity and degree calculations for the six families, and a finite flag identity. Taking consecutive differences gives the individual coefficients.

On the diagonal, their highest-weight module agrees with ours after scalar extension. In higher degrees they obtain exact rational and quantum cohomology dimensions and lower bounds for finite groups. Section 6 records the transfers and their hypotheses.

For $m \geq 3$, the normalized completion and the simple-reflection chains give different canonical expansions. The completion identifies the two uncorrected products exactly. The chains introduce lower finite canonical summands, which we describe along with the corrections in both calculations.

Sources and notation

A denotes our paper *Annular webs and Guralnick's conjecture*, arXiv:2611.09114v1.

B denotes their paper *The cohomology of finite simple groups is unbounded*, arXiv:2611.11677v1.

We use the numbering of these two versions. Section 4, pages 9–10, gives the proof of the individual formula. The `comparison/` directory in our repository contains the source of this note, both Python implementations and their recorded outputs. Our Lean development verifies the row sum in A.

1 Conventions and the common canonical element

Put $r = m,\ N = 2m+1,\ I = \{s_1, \dots, s_{N-1}\},\ w_I = w_0$, and $V = I \setminus \{s_m, s_{m+1}\}$. Both papers compose periodic permutations by $(xy)(i) = x(y(i))$. Right multiplication by s_i swaps positions $i, i+1$. For s_0 the new first and last entries are $w(N) - N, w(1) + N$. The increasing window selects the shortest representative in W/W_I; the increasing inverse window selects the shortest representative in $W_I \backslash W$. We work in the extended group. Our construction has degree zero.

Parameters. Both use $H_s^2 = 1 + (v^{-1} - v)H_s,\ C_s = H_s + v$. With $T_w = v^{-\ell(w)} H_w$, the coefficient of H_y in C_w is $h_{y,w}(v) = v^{\ell(w)-\ell(y)} P_{y,w}(v^{-2})$. We use $q = v^{-2}$ in our flag counts. Bowman, de la Fuente, Libedinsky and Plaza use $q = v^2$ for their Gaussian polynomials (B, §3.1). Gaussian reciprocity contributes the factor $v^{-2b(a-b)}$ when changing $\begin{bmatrix} a \\ b \end{bmatrix}_{v^{-2}}$ to $\begin{bmatrix} a \\ b \end{bmatrix}_{v^2}$. For $L = \ell(w_P)$, we write C_P, h_P for their C_{w_P}, π_P, respectively:

$$C_P = \sum_{z \in W_P} v^{L-\ell(z)} H_z, \qquad h_P = \sum_{z \in W_P} v^{L-2\ell(z)}, \qquad C_P^2 = h_P C_P.$$

Sides and bases. Our web maps the right ideal $C_I\mathcal{H}$ to $C_V\mathcal{H}$. Evaluate on C_I, then apply the linear anti-involution $H_w^* = H_{w^{-1}}$, which fixes v and reverses products. The result lies in their right ideal $C_I\mathcal{H}$, with right V-spherical character. For a double coset $D = W_V d W_I$ with longest element z, its standard map goes to $v^{\ell(z)} \sum_{w \in D^{-1}} T_w$, and its canonical map to $C_{z^{-1}}$. Their bases are $M_b = C_I H_b$ and $\mathbf{C}_b = C_{w_I b}$, for $b \in {}^I W$, and $[M_a]\mathbf{C}_b = h_{w_I a, w_I b}$. At our splitter the inverted standard is $M_e = C_I$. Thus for *all* j, with $b = u_{j;m}^{-1}$,

$$\big[[B_m]_v\big]\theta_{A_{j;m}} = [M_e]\mathbf{C}_b = v^{D_{j;m}} P_{w_0, C_{j;m}}(v^{-2}), \qquad D_{j;m} = 2j(N-j) - 2(m-j) - 3.$$

Here $C_{j;m} = u_{j;m} w_0$ is a permutation, while C_w denotes a basis vector. See A Definitions 2D.1–3, 2E.1–2, Lemma 2E.7, Proposition 2E.9 and §3B; B §2.1, Lemma 2.1.

The diagonal. Write $\nu = (1^m, 0, (-1)^m)$, $\sigma = (m, m+2)$, and $t_\nu(i) = i + N\nu_i$ on $1 \le i \le N$. In our increasing window at $j = m$ the ordered residues are $1, \dots, m-1, m+2;\ m+1;\ m, m+3, \dots, N$ with shifts $-1, 0, 1$ on the respective groups. Therefore $u_{m;m} = \sigma t_{-\nu}$. Its inverse $t_\nu \sigma$ has window $[N+1, \dots, N+m-1, 1-m, m+1, N+m, 2-m, \dots, 0]$, the window of their element β_m. As $w_I \beta_m = C_{m;m}^{-1}$, inversion identifies the ordinary KL polynomials and the full Hecke vectors satisfy $(\theta_{A_{m;m}}(C_I))^* = \mathbf{C}_{\beta_m}$. The dot action gives $\beta_m \cdot_p 0 = p\nu + \sigma\rho - \rho = p\nu - 2(\alpha_m + \alpha_{m+1})$, hence

$$\lambda_{m;m}(p) = 2\varpi_{m-1} + (p-2)(\varpi_m + \varpi_{m+1}) + 2\varpi_{m+2}.$$

They include the factor p in the translation label. See A Lemma 6A.2 and B Lemma 2.16.

2 The two galleries in one Schur algebroid

Write $C_P = C_{w_P}$, $\pi_P = h_P$, $\Pi_n = v^{-\binom{n}{2}}[n]_{v^2}!$, $[n] = \Pi_n/\Pi_{n-1}$, and $\eta = [2] = v + v^{-1}$. For a consecutive interval $[a, b]$ in a parabolic set we mean $s_a, \dots, s_b$. In the right spherical module the sequence of merge targets $P_1, \dots, P_d$ represents

$$D(P_1, \dots, P_d) = \frac{C_I C_{P_1} \cdots C_{P_d}}{\prod_{i=1}^{d} \pi_{P_{i-1} \cap P_i}}, \qquad P_0 = I.$$

Between consecutive targets one splits to their intersection and merges to the next target. The splits are inclusions. Before inversion the merge from X to $Y \supset X$ is left multiplication by C_Y/π_X on the appropriate image. Inversion reverses the product. Thus every quotient below is integral on its specified domain (A, Lemma 3B.1; B, Definition 3.2 and Lemma 3.3).

Symbol	Generators	Position blocks or symmetric-group type
I	$[1, N-1]$	(N)
J	$S \setminus \{s_m, s_{m+1}\}$	$(2m, 1)$ across the cut
V	$I \cap J$	$(m, 1, m)$
K	$[2, N-2]$	$(1, 2m-1, 1)$
L	$K \cap J = K \cap V$	$(1, m-1, 1, m-1, 1)$
K'	$I \setminus \{s_1, s_{m+1}\}$	$(1, m, m)$
E	$[3, N-1]$	$(1, 1, 2m-1)$
T	$J \cap K' = V \cap K'$	$(1, m-1, 1, m)$
D'	$E \cap K'$	$(1, 1, m-1, m)$

In particular $\mathcal{R}_m = D(J, K, V)$ and $\mathcal{BS}_m = D(J, K', E)$. Both start with $F = C_I C_J/\pi_V$; this is itself canonical by B, Lemmas 3.1 and 3.19. The first two stages of our web are $I \supset V \subset J$. For the general $\mathcal{BS}_j$ replace J, K' by $J_j = S \setminus \{s_j, s_{j+1}\}$ and $K_j = I \setminus \{s_1, s_{j+1}\}$. The cut is crossed by merging the final $N-j-1$ packet with the first j packet; the singleton stays separate. The intermediate splittings have blocks $(j, 1, N-j-1)$, $(1, j-1, 1, N-j-1)$, and $(1, 1, j-1, N-j-1)$, apart from the degeneracy at $j = 1$, where the last two overlaps both equal $K_1 = E$. For $\mathcal{CBS}_s = D(P_s, E)$, $P_s = S \setminus \{s_s\}$; one joins the packets $(s, N-s)$ across the cut, then splits to $(1, 1, s-2, N-s)$ and merges the last two packets, omitting zero packets. These rotations and their inverse undo one another in determinant degree. Explicitly $C_{J_j} = \tau^{-(N-j-1)} C_{[1,N-2]} \tau^{N-j-1}$ and $C_{P_s} = \tau^{-(N-s)} C_I \tau^{N-s}$ (A, Lemma 2C.9).

For $j \geq 2$ the three overlaps for $\mathcal{BS}_j$ have factors $\Pi_j \Pi_{N-j-1}$, $\Pi_{j-1} \Pi_{N-j-1}$, $\Pi_{j-1} \Pi_{N-j-1}$; for $j = 1$ these same scalar formulas hold. For $\mathcal{CBS}_s$ they are $\Pi_s \Pi_{N-s}$ and $\Pi_{s-2} \Pi_{N-s}$. Our three factors are $\Pi_m^2, \Pi_{m-1}^2, \Pi_{m-1}^2$. The length contribution at an individual merge is $\ell(w_{P_i}) - \ell(w_{P_{i-1} \cap P_i})$.

An exact identity before canonical correction

Set $Q_1 = [2, m]$, $Q_2 = [1, m-1]$ and let T_1, T_2 denote right multiplication by $C_{Q_1}/\Pi_{m-1}, C_{Q_2}/\Pi_{m-1}$, respectively, on the indicated spherical subspaces. Then

$$\boxed{\mathcal{R}_m = D(J, K', E, K', V) = \mathcal{BS}_m T_1 T_2.} \qquad (*)$$

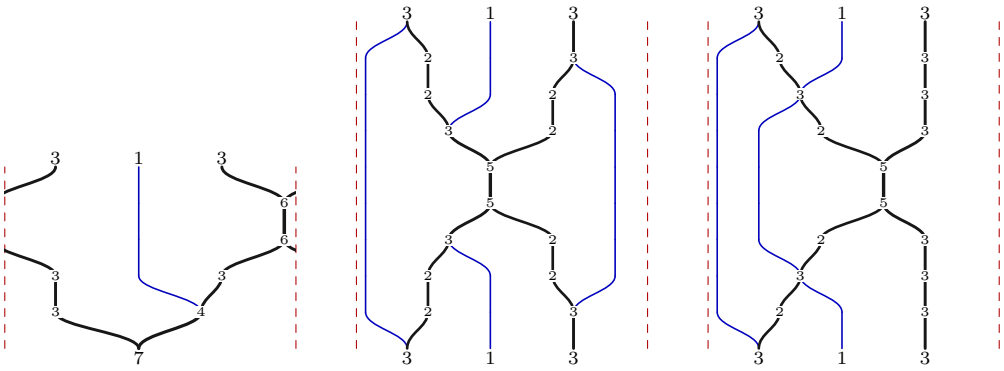
The maximal moving words in these two normalized parabolic factors are exactly $s_2 \cdots s_m$ and $s_1 \cdots s_{m-1}$ from B, Lemma 2.4. The equality holds for the full standard and canonical expansions.

Proof. Refine J to V in both galleries. The first overlaps are $L \subset V$ and $T \subset V$, so this factors an inclusion. Everything after this refinement is a finite flag calculation. Fix the two end flags $U \subset U'$ and $W \subset W'$ of dimensions $m, m+1$ in an N-space over $\mathbb{F}_q$. In the V, K, V gallery the K-flag is a line $a \subset U \cap W$ and a hyperplane $H \supset U' + W'$. In the V, K', E, K', V gallery the two K'-flags are $(a, U'), (a, W')$, and the E-flag is (a, D_2) with $a \subset D_2 \subset U' \cap W'$, $\dim D_2 = 2$. All refinements are uniquely determined by these data. If $d = \dim(U' \cap W')$, then

$$\dim((U' \cap W')/a) = d - 1 = N - \dim(U' + W').$$

For each a, the numbers of two-planes and hyperplanes are therefore equal. Each gallery counts $[\dim(U \cap W)]_q[d-1]_q$ points, where $[k]_q = 1 + \cdots + q^{k-1}$ and $[0]_q = 0$.

The normalization shifts agree. Writing $b_n = \binom{n}{2}$, the first tail contributes $b_{2m-1} + 2b_m - 4b_{m-1}$, and the second contributes $b_{2m-1} + 6b_m - 4(b_m + b_{m-1})$. The two galleries have the same input and output flags, so the orbit-dependent standard shifts also agree. Equality over every finite field gives the generic Schur identity. Finally $K' = Q_1 \times [m+2, N-1]$ and $V = Q_2 \times [m+2, N-1]$. The common factor lies in E, commutes with Q_1, Q_2, and cancels one Π_m from each of $\pi_{D'} = \pi_T = \Pi_m \Pi_{m-1}$. This gives $(*)$. □



Common affine part Line and hyperplane Line and plane

Cut-open webs at $m = 3$, read upwards. Stack the common part below either tail and identify each pair of red dashed sides. Blue strands have label one. The label-six strand in the common part straddles the cut. The vertices are the normalized splits and merges of A 2D.4.

The product of the $2m-2$ individual simple KL elements differs from the normalized completion for $m \geq 3$. At the identity coordinate, multiplication by $T_1 T_2$ acts by $[m]^2$, whereas the individual factors act by η^{2m-2} (B, (3.4)). Since $\varepsilon(\mathcal{BS}_m) \neq 0$, evaluation at $v = 1$ distinguishes the two products by $m^2 \neq 4^{m-1}$ for $m \geq 3$. Averaging further by C_V/π_V preserves this coordinate. At $m = 2$ each parabolic factor is already one simple KL element, so the raw products agree.

Both finite tails are canonical

The finite tails can be compared in finite type A before applying the common affine prefix. In the finite spherical module $C_V\mathcal{H}(S_N)$ put

$$\mathsf{a}_m = \frac{C_V C_K C_V}{\pi_L^2}, \qquad \mathsf{b}_m = \frac{C_V C_{K'} C_E}{\pi_{V\cap K'}\pi_{K'\cap E}}.$$

The right V-eigenrelations of $F = C_I C_J/\pi_V$ define the module map

$$\Phi : C_V\mathcal{H}(S_N) \longrightarrow C_I\mathcal{H}(W), \qquad \Phi(C_V h) = Fh.$$

Thus $\Phi(\mathsf{a}_m) = \mathcal{R}_m$ and $\Phi(\mathsf{b}_m) = \mathcal{BS}_m$, with all normalization factors included. Both finite tails are single ordinary canonical elements:

$$\mathsf{a}_m = C_{w_A}, \qquad w_A = w_0 w_{\{s_m, s_{m+1}\}}, \qquad \mathsf{b}_m = C_{w_B}, \qquad w_B = [m, m+1, N, N-1, \ldots, m+2, m-1, \ldots, 1].$$

Their minimal left V-representatives have lengths m^2+2m-3 and m^2-1, respectively.

Proof. In the three-letter notation of the parabolic paper, let $c_0 = c_{\mathsf{A}^m\mathsf{BC}^m}$, with window $[1-m, \ldots, 0, m+1, N+1, \ldots, N+m]$ and length m^2. For $x \in {}^V W_I$, the map $C_V H_x \mapsto M_{c_0 x}$ identifies the finite standard basis with the type-m standard span. The window formula gives $\ell(c_0 x) = m^2 + \ell(x)$, and the equal-letter and unequal-letter multiplication rules show that it commutes with finite right Hecke actions (B, Lemmas 3.8–3.9). The type-m part of F is just M_{c_0} with coefficient one (B, Lemma 3.19). Consequently the standard coefficients of the two finite tails are exactly the type-m coefficients of the two affine products.

At the boundary $m - r = 0$, our six-family estimates give coefficient one at the unique top and strictly positive valuation at every proper orbit maximum; refinement adds the nonnegative length deficit (A, Lemmas 4C.2 and 4D.1). The lowest-exponent table in the parabolic paper at type m has a unique AB orbit maximum of valuation zero; the AA maxima have valuation one, and every nonmaximum has an additional positive length deficit (B, Proposition 3.21). The top coefficient is one in each case. The two top words are $\mathsf{C}^{m-1}\mathsf{ABCA}^{m-1}$ and $\mathsf{ABC}^m\mathsf{A}^{m-1}$, which give the displayed finite permutations. Integrality and bar invariance of the normalized products now identify both tails by canonical triangularity. □

Let $L_m = C_{s_2}\cdots C_{s_m} C_{s_1}\cdots C_{s_{m-1}}$. The corresponding reflection word takes w_B to w_A, increasing length at every step: first it moves $m+1$ past the larger entries, then it moves m past them. Finite KL multiplication therefore gives

$$\mathsf{b}_m L_m = \mathsf{a}_m + \sum_{z<w_A} d_z(v) C_z, \qquad d_z(v) \in \mathbb{N}[v, v^{-1}],$$
$$\mathcal{BS}_m L_m = \mathcal{R}_m + \sum_{z<w_A} d_z(v)\Phi(C_z).$$

Only finite canonical indices with left V-descents occur in the sum. The top multiplicity is one. This expresses our web as the image of the top finite canonical summand after the two simple-reflection chains. Using the normalized completion instead gives the exact finite identity $\mathsf{b}_m T_1 T_2 = \mathsf{a}_m$.

For $m = 3$, the correction has the following form. On the appropriate spherical domain, the S_3 identity $C_{\{s,t\}} = C_s C_t C_s - C_s$ gives

$$\mathcal{R}_3 = \mathcal{BS}_3(C_2C_3 - 1)(C_1C_2 - 1), \qquad C_i = C_{s_i}.$$

For $m = 2, 3, 4, 5$ the uncorrected finite product $\mathsf{b}_m L_m - \mathsf{a}_m$ has respectively $0, 3, 10, 20$ distinct canonical terms. These are exact finite checks; the canonical-tail and summand statements above hold for all m.

What the corrected seed contains

Write $b_j = u_{j;m}^{-1}$ and let $[\,\mathbf{C}_b\,]$ denote a coefficient in the spherical canonical basis. For every $1 \le j \le m$,

$$\boxed{[\,\mathbf{C}_{b_j}\,](\mathbf{C}_{X_0}T_1T_2) = \delta_{jm}.} \tag{$**$}$$

At the other end of $(*)$, B, Proposition 3.21, also gives $\mathcal{BS}_m = \sum_{j=1}^{m} \mathbf{C}_{z(j,1,0)}$. In fact its standards are in $\mathbb{Z}[v]$, with constant term 1 precisely at the m indicated orbit maxima and zero elsewhere. Subtracting those canonical elements leaves a bar-invariant vector in $v\mathbb{Z}[v]\{M_c\}$, hence zero. Thus $\mathcal{R}_m = \sum_{j=1}^{m} \mathbf{C}_{z(j,1,0)}T_1T_2$. In particular $(**)$ and A, Lemma 4I.1, place each off-diagonal family-F term, once, in

$$(\mathcal{BS}_m - \mathbf{C}_{X_0})T_1T_2 = \Big(\mathcal{BS}_{m-1} - \eta \sum_{s=2}^{m-1} \mathcal{CBS}_s\Big)T_1T_2.$$

Here $\mathcal{BS}_{m-1}$ is taken in the same period $2m+1$.

Proof of $(**)$. For consecutive generators starting at s_a let $Z_0 = Z$, $Z_i = ZC_{[a,a+i-1]}/\Pi_i$; assume $ZH_s = v^{-1}Z$ for $s \in [a+1, a+i-1]$. The left coset representatives are $g_k = s_a \cdots s_{a+k-1}$, $0 \le k \le i$, so that

$$Z_i = \sum_{k=0}^{i} v^{i-k} Z H_{g_k}, \quad Z_1 = ZC_{s_a}, \quad Z_i = Z_{i-1}C_{s_{a+i-1}} - Z_{i-2} \quad (i \ge 2).$$

The last generator commutes with g_k for $k \le i-2$, giving the last recurrence. In B, Proposition 2.8, this subtracts the preceding cover at each step. The remaining terms $\mathbf{C}_e, \mathbf{C}_{\gamma_k}$ are eigenvectors for every later generator, since the word for γ_k is $BC^kB^{N-2k-1}A^k$. Using $[i+1] = \eta[i] - [i-1]$ gives

$$\mathbf{C}_{X_0}T_1 = \mathbf{C}_{x_{m-1}} + [m-1]\mathbf{C}_e + \sum_{k=1}^{m-2} [m-1-k]\mathbf{C}_{\gamma_k}.$$

The terms after $\mathbf{C}_{x_{m-1}}$ retain the right descent s_{m+1} upon multiplication by T_2, as s_{m+1} commutes with Q_2. Applying the same recurrence to B, Proposition 2.11, yields

$$\mathbf{C}_{x_{m-1}}T_2 = \mathbf{C}_{\beta_m} + \sum_{t=0}^{m-2} [t+1]\mathcal{E}_t.$$

Indeed each canonical term of $\mathcal{E}_t$ is sorted on its first $m+1$ positions by B, Lemma 4.12(ii) and (4.17)–(4.18). Every subsequent generator acts by η, and each such term has right descent s_m. The canonical basis of the right s-spherical subspace consists of the canonical elements with right descent s. The word of b_j is $C^{j-1}B^{m-j}A \,|\, B \,|\, CB^{m-j}A^{j-1}$. The adjacent ascents AB, BC give both missing right descents s_m, s_{m+1} (B, Lemma 4.9). Therefore none of the remainders contains any $\mathbf{C}_{b_j}$. □

By B, Proposition 2.6, the identity coordinate of $\mathbf{C}_{X_0}T_1T_2$ has valuation $3-2m$. For $\mathbf{C}_{\beta_m}$ the valuation is one, so the full vectors differ. Formula $(**)$ isolates the diagonal within family F. Section 4 computes the degree-one identity coordinate of each family-F constituent by completing successive partial sums.

3 Two ways of organizing the coefficient calculation

Both arguments use the same refinement identity. If z is the longest index in a right-parabolic orbit, the coefficient of M_c in a vector invariant under that parabolic is $v^{\ell(z)-\ell(c)}$ times its coefficient at z. One proves this along a single edge by the quadratic Hecke relation. We use the identity to refine output flags (A, Lemma 4D.1). Bowman, de la Fuente, Libedinsky and Plaza use it to compute once per orbit (B, Lemmas 3.11, 3.17 and Proposition 3.21).

Two multiplicity-one expansions. The uncorrected product has a multiplicity-one expansion. Write $z(t,a,b)$ for the longest element of their orbit of type t with indicators a,b for the first two letters. Then

$$\mathcal{BS}_m = \sum_{t=1}^{m} \mathbf{C}_{z(t,1,0)}, \qquad \mathcal{BS}_{m-1} - \eta \sum_{s=2}^{m-1} \mathcal{CBS}_s = \sum_{t=1}^{m-1} \mathbf{C}_{z(t,1,0)}, \qquad \eta = v + v^{-1}.$$

Indeed, B Proposition 3.21 gives least exponents $1,1,0,1$ at feasible maxima with prefixes BB, BA, AB, AA, respectively, for $\mathcal{BS}_m$. The AB maxima occur exactly for $1 \le t \le m$, and their constant terms are one. All other standard coordinates have positive valuation by refinement. Subtract the canonical vectors displayed in the first sum. The difference is bar invariant and every standard coordinate lies in $v\mathbb{Z}[v]$, so it vanishes (B Lemma 3.1). B Proposition 3.5, with $X_0 = z(m,1,0)$, gives the second equality. In its proof the AB constants below the top cancel against $\mathcal{BS}_{m-1}$; in column AA the $-v^{-1}$ from that subtraction is canceled by $+\eta\mathcal{CBS}_t$ at each type $2 \le t < m$. This isolates a single seed.

We prove multiplicity one for the completed web. In A, Lemma 3C.3, we fix flags $I_0 \subset K_0$ of dimensions $r, N-r$ and $F \subset F'$ of dimensions $m, m+1$. For a line $L \subset F$ and a hyperplane $H \supset F'$, put $e = \mathbf{1}_{L \not\subset I_0}$, $d = \mathbf{1}_{H \not\supset K_0}$. The remaining choices of $I' \subset K'$ of dimensions $m, m+1$, between $I_0 + L$ and $K_0 \cap H$, number $\begin{bmatrix} N-2r-e-d \\ m-r-e \end{bmatrix}_q (m+1-r-d)_q$ over a field of size q. Sum against the corresponding line and hyperplane counts (A (3C.2)), substitute $q = v^{-2}$ and multiply by v^{D-d_A}, $D = 2m^2 + 2m - 3$ (A Proposition 3C.4). If $F_A(q)$ is the resulting counting polynomial before normalization, the exponents of a nonzero canonical multiplicity lie between $-E$ and E, $E = 2\deg F_A - (D - d_A)$. We apply positivity after clearing the overlap denominator. With $l = m - r$, the bounds for the six allowed families A–F are respectively $2l-3, l-2, l-1, l-2, l-1, 0$ (Lemmas 4B.1 and 4C.2). At an F-standard $A_{j;m}$, a full-spherical, strong one-incidence or weak one-incidence contribution of rank $j+h$ starts in degree at least $2h+6, 2h+2, 2h+4$, respectively. Larger F-terms contribute only positive powers there. Thus the constant one belongs to $A_{j;m}$ itself (Lemma 4I.1). At a full-spherical standard, in contrast, one-incidence terms with their multiplicities start in degree at least three (Lemma 4H.1), which suffices for the row sum.

The same Gaussian and different subtractions

The common Gaussian has a direct flag-counting explanation. At the zero orbit, our gallery chooses $L \subset I' \subset K' \subset H$ of dimensions $1, m, m+1, 2m$, with $L \subset F$ and $F' \subset H$, where $\dim F = m$ and $\dim F' = m+1$. There are $(m)_q$ choices each for L and H. In the $(2m-1)$-space H/L, the remaining flag $I'/L \subset K'/L$ contributes $\begin{bmatrix} 2m-1 \\ m-1 \end{bmatrix}_q (m)_q$. The full count is therefore $(m)_q^3 \begin{bmatrix} 2m-1 \\ m-1 \end{bmatrix}_q$.

Let $t = v^2$, $G = \begin{bmatrix} 2m-1 \\ m-1 \end{bmatrix}_t$, and use $(k)_t = 1 + \cdots + t^{k-1}$, $[k] = v^{1-k}(k)_t$. At the identity the normalized raw values are

$$\epsilon(\mathcal{R}_m) = v^{3-2m}(m)_t^3 G, \qquad \epsilon(\mathcal{BS}_m) = v(m)_t G.$$

These instances of the same rectangle follow from the operator identity and the finite eigenvalue, with ratio $[m]^2$ (A Prop. 3C.4, B proof of Prop. 2.6 in §3.5). The auxiliary operators satisfy related identities after merging all outputs. Define $U_j = C_I C_{J_j} C_I / \pi^2_{I \cap J_j}$ and let $V_s = C_I C_{P_s} C_I / \pi^2_{I \cap P_s}$ be our spherical web (Definition 4E.1). With $p_E(F) = FC_I/\pi_E$,

$$p_E(\mathcal{BS}_j) = [j]^2 U_j, \qquad p_E(\mathcal{CBS}_s) = [s][s-1]V_s.$$

Indeed $C_Q C_I = \pi_Q C_I$ for $Q \subset I$. The two factorial ratios remaining are $\pi_{I \cap J_j}\pi_{K_j}/(\pi_{J_j \cap K_j}\pi_{K_j \cap E}) = [j]^2$ and $\pi_{I \cap P_s}/\pi_{P_s \cap E} = [s][s-1]$. Thus the parabolic seed correction includes scalar multiples of our cumulative spherical webs after the total merge. If instead one merges first to $Q = I \setminus \{s_1\}$ and appends the move to $I \setminus \{s_f\}$ for $f = m, m+1$, the result for $\mathcal{CBS}_s$ is $[s-1]$ times our one-line gallery $\mathcal{E}_{s,f}$. This follows from $\pi_{P_s \cap Q}/\pi_{P_s \cap E} = [s-1]$ and A Definition 4F.1.

We take the differences $C_{Z_s} = V_s - V_{s-1}$, whose standard coefficients are differences of consecutive Gaussians (Lemma 4E.4). We subtract $g_{m-s} C_{Z_s}$, where $g_l = \sum_{a=0}^{l-2} \binom{a+3}{3}(v^{2l-3-2a} + v^{-(2l-3-2a)})$. To determine these multiplicities, we invert a convolution of Gaussian polynomials and truncate modulo q^m; the common series $1/((1-q)^3(q;q)_\infty)$ cancels, leaving $\binom{m+1}{2} q^{m-1}$ (Lemmas 4J.1–4K.1). The surviving linear coefficient at the origin is the sum over family F (Theorem 4L.1).

In the parabolic paper the authors apply ϵ to their corrected seed to obtain $h_{X_0} = vG + O(v^{2m+1})$. The first m coefficients are ordinary partition numbers, since partitions of size below m fit the Gaussian rectangle. The first reflection chain satisfies $h_{x_1} = \eta h_{X_0} - 1$ and

$$h_{x_{k+1}} = \eta h_{x_k} - h_{x_{k-1}} - v^k \begin{bmatrix} N-1 \\ k \end{bmatrix}_t \quad (1 \le k \le m-2).$$

The last term is h_{γ_k} (Lemma 3.25). Bruhat covers supply the predecessor; positivity excludes the lower γ_j, paired deletion reduces the surviving multiplicity to one, and a nonidentity seed bound excludes the κ_j (Proposition 2.8, Lemmas 3.24, 4.20, 5.1). If $d_{a,b}$ counts partitions of $a+b$ with at least a parts, the coefficient of v^{a+1+2b} in h_{x_a} is $\sum_{i=0}^{a} d_{i,b}$ for $0 \le b \le m-1-a$ (Lemmas 2.9, 5.2). In particular $h_{x_{m-1}}$ starts with mv^m.

On the second chain the predecessor and remainder have identity-coordinate valuation at least $t'+3$ in the step to $y_{t'}$ (Proposition 2.11 and Lemma 2.12). Of its noncover candidates the ordered terms vanish after rank reduction, the pure terms have sufficiently high valuation by finite ascent bounds, and mixed terms vanish after a rank-two star operation and a descent test (Lemmas 6.1, 6.2, 6.5). This transports m down one degree at each step. They illustrate the remainders by explicit computation at $m = 5$ and prove the degree bound in every rank. Their partition recurrence follows the coefficient along a chain. Our Gaussian convolution computes the row sum from the six-family base change. The completion identity relates the products underlying these two calculations.

Relation to Scott–Xi. Scott–Xi, arXiv:0909.3394v4, Theorem 3.3, obtained $n+2$ for $\mu(xw_0, vxw_0)$ with a regularly translated lower endpoint. For the word in their Lemma 3.4 the fixed-source coefficient is 2 in every rank $n \ge 4$. At $n = 4$ this gives the $m = 2$ example in both papers. Scott–Xi pose the fixed-source question in Section 4. Their suggested weight for $n = 4k$ has coefficient $p-2$ at every internal node, so for $n > 4$ it differs from the common diagonal weight considered here. The low-rank coincidence, the fixed-source question and the use of degree bounds in a KL recursion all have this earlier precedent.

4 Every entry of the row

Combining our six-family bounds with the parabolic orbit estimates in B and applying the normalized completion gives

$$\boxed{\mu_{j;m} = j \quad (1 \le j \le m).}$$

The proof takes the first j canonical summands of $\mathcal{BS}_m$ and applies T_1T_2, with the ambient rank and our output boundary fixed.

1. Nested canonical sums before completing the web. Hereafter val denotes the lowest exponent of v, with $\operatorname{val} 0 = +\infty$. Let $T = T_1T_2$ be the normalized finite tail which gives $\mathcal{BS}_m T = \mathcal{R}_m$. We use their notation for the longest orbit representatives and put

$$X_i = \mathbf{C}_{z(i,1,0)}, \qquad D_s = \mathbf{C}_{z(s,1,1)}, \qquad \Phi_j = \sum_{i=1}^{j} X_i, \qquad Y_j = \Phi_j T, \qquad Y_0 = 0.$$

Here X_i is a vector. The lower-case x_i in the parabolic paper is a permutation on a reflection chain. We show

$$\mathcal{BS}_j = \Phi_j + \sum_{s=2}^{\min(j,m-1)} a_{js}(v) D_s, \qquad D_s = \mathcal{CBS}_s, \qquad \operatorname{val} a_{js} \ge 2 - N + 2j \quad (a_{js} \ne 0). \tag{1}$$

Only $s \le m-1$ is needed: at $j = m$ every $a_{js} = 0$. For $s \le m-1$ B Proposition 3.21 gives $\mathcal{CBS}_s$ a unique coordinate with nonpositive valuation: at the AA maximum of type s, its constant is one. All others are positive, including nonmaxima by refinement. This proves its canonicality. The $\mathcal{BS}_j$ are right E-invariant, so only maxima of E-orbits can be canonical indices; this is the parabolic canonical basis, or follows by descending triangularity in the invariant subspace. Outside the support listed in B Proposition 3.21 no canonical term can occur: clearing the Poincaré denominator gives a positive product in the ordinary KL basis. Apply the same valuation argument as A Lemma 4B.1 at the maxima. Columns BB, BA have positive valuation. Column AB has valuation zero, so its canonical multiplicity is constant. Column AA gives the bound $2 - N + 2j$ in (1). At an AB maximum of type i the coordinate of $D_s = \mathcal{CBS}_s$, when present, has valuation $N - 2i$ by the same table. Multiplied by a_{js} it starts no lower than $2 + 2(j - i)$. Other AB canonical vectors with constant multiplicities contribute only positive powers there. Thus the constant one belongs to X_i for every feasible $1 \le i \le j$. This proves (1) and in particular $Y_m = \mathcal{R}_m$.

2. Support and multiplicity bounds. Each Y_j is integral and bar invariant. The domain of T is the right E-invariant module; both normalized factors are integral sums over finite parabolic cosets there (B Lemma 3.3). If $p = (\Pi_{m-1})^2$ is their common denominator, $p(Y_j - Y_{j-1}) = X_j C_{Q_1} C_{Q_2}$ has nonnegative ordinary canonical coefficients. Consequently the cleared coefficient of *each* Y_j is bounded coefficientwise by that of $p\mathcal{R}_m$. Thus every canonical index of Y_j belongs to one of our six families. The valuation of a nonzero multiplicity is at least the corresponding valuation for $\mathcal{R}_m$, by subtracting $\operatorname{val} p$ from the inequality for cleared coefficients. This proves that the degree margins in A Lemmas 4H.1 and 4I.1 apply to Y_j as well. In particular, for each family-F index its multiplicity in Y_j is a constant zero or one, nondecreasing in j, since its multiplicity in $\mathcal{R}_m$ is one. All standard and canonical indices of Y_j have type at most j: (1) and B Proposition 3.21 bound the initial standard support, finite right generators preserve word type, and positivity after clearing excludes any additional canonical support.

For completeness we compute the constant at $A_{j;m}$. A flag for $\mathcal{BS}_j T$ chooses intermediate $U \subset U'$ of dimensions $j, j+1$, with $I_0 \subset U \subset U' \subset K_0$, in the finite quotient of A Definition 3C.1. To check the input conditions directly, B Lemma 3.19 restricts the first parabolic product to words whose A letters are in the first j positions, whose singleton is neutral and whose C letters are after position $j+1$. On the rank orbit this says precisely $I_0 \subset U \subset U' \subset K_0$. The one-cut product similarly imposes just $I_0 \subset U \subset K_0$. Equivalently, divide the nested periodic lattices by the final lattice as in our proof of Proposition 3C.4 in A. For the output $F \subset F'$ of dimensions $m, m+1$, it also chooses a line $L \subset U \cap F$ and a two-plane D with $L \subset D \subset U' \cap F'$. Indeed the flags between the first affine parabolic and the output successively retain (L, U'), (L, D), (L, F'); this works also when $j = 1$ with the repeated cut suppressed. For $\mathcal{CBS}_s T$, the first affine parabolic retains just an s-space U with $I_0 \subset U \subset K_0$, and the final conditions become $L \subset F$, $L \subset D \subset U \cap F'$.

At $A_{j;m}$ the rank is j, so $U = I_0$ in the first gallery. A Lemma 4C.2 gives $\dim(I_0 \cap F) = \dim(I_0 \cap F') = 1$ and $\dim(K_0 \cap F') = m - j + 2$. Thus L is forced. The line U'/I_0 must belong to the image of $K_0 \cap F'$ modulo I_0; then $U' \cap F'$ is the unique possible D. The count is $(m-j+1)_q$ where q is the flag parameter. Its normalization shift is $2(m-j)$: the gallery normalization constant is

$$d_j^{\mathrm{BS}} = 4m - 3 + 2(j-1)(N-j-1),$$

as follows either by summing the parabolic length shifts or by the origin count below, and A Lemma 4C.2 gives $d_{A_{j;m}} = 2j(N-j) - 2(m-j) - 3$. Thus the normalized coordinate has constant one. For $\mathcal{CBS}_s T$, $s < j$ is impossible and $s = j$ would require $D \subset I_0 \cap F'$, also impossible. The constant therefore belongs to Y_j. By the margins of A Lemma 4I.1 it is the multiplicity of $C_{A_{j;m}}$. Monotonicity and the type bound now show that the family-F indices occurring in Y_j are exactly $A_{1;m}, \ldots, A_{j;m}$, once each.

3. Coordinates at the full-spherical standards. At Z_r we have $I_0 \cap F' = 0$, $\dim(F \cap K_0) = m - r$, and $\dim(F' \cap K_0) = m - r + 1$, by A Definition 3C.1 and Lemma 4C.2. The choices of L, D in K_0 number $(m-r)_q^2$, and D is disjoint from I_0. For the BS gallery choose U'/I_0 containing the image of D, then a hyperplane U/I_0 containing the image of L. For the CBS gallery choose U/I_0 containing the image of D. The two flag counts are, respectively,

$$(m-r)_q^2 (j-r)_q \begin{bmatrix} N-2r-2 \\ j-r-1 \end{bmatrix}_q, \qquad (m-r)_q^2 \begin{bmatrix} N-2r-2 \\ s-r-2 \end{bmatrix}_q.$$

The first vanishes for $r \geq j$, the second for $r > s-2$; these cases are omitted before evaluating Gaussians. The standard normalization is $v^{d-d_{Z_r}}$ times the count at $q = v^{-2}$, where $d_{Z_r} = 2r(N-r)$ in our normalization. To verify the absolute shifts, use the origin formulas in B, proof of Proposition 2.6, multiplied by $[m]^2$ for T. Reversing the displayed counting polynomials at $r = 0$ gives $d = d_j^{\mathrm{BS}}$ as above and $d = d_s^{\mathrm{CBS}} = 6m - 2s + 2(s-2)(N-s)$. The flag normalization depends on the endpoint standard only through d_{Z_r} (A Lemmas 2D.4, 2E.3). Subtracting $2r(N-r)$ in either formula amounts to replacing (m, j, s, N) by $(m-r, j-r, s-r, N-2r)$.

Set $h = j - r > 0$, $t = v^2$. On its support the BS coordinate is therefore

$$v^{3-2h} (m-r)_t^2 (h)_t \begin{bmatrix} 2(m-r)-1 \\ h-1 \end{bmatrix}_t.$$

The corresponding CBS coordinate has valuation $2m + 4 - 2s$; by (1) its contribution to Y_j starts no lower than $5 + 2(j-s)$. Consequently, through degree one inclusive, the Z_r-coordinate of Y_j agrees with the Z_r-coordinate of $\mathcal{R}_j$ in *ambient parameter* j. Indeed $(m-r)_t \equiv (h)_t \bmod t^h$; both Gaussian numerators in the rectangle product formula start contributing only in degree h or higher. Multiplication by v^{3-2h} puts the error in degree at least three. For $r \geq j$ these coordinates vanish.

4. The same spherical corrections in the smaller parameter. For $r \leq s \leq j$, the Z_r-coordinate of the canonical C_{Z_s} is

$$\begin{bmatrix} 2(m-r)+1 \\ s-r \end{bmatrix}_t - \begin{bmatrix} 2(m-r)+1 \\ s-r-1 \end{bmatrix}_t$$

(A Lemma 4E.4). It agrees modulo t^{j-r+1} with the same coordinate in ambient parameter j: in both rectangle formulas the first numerator exponent is at least $j - r + 2$, unless the lower entry is negative, in which case the Gaussian is zero. In particular, for $s \leq j - 2$ this agreement persists through degree one after multiplication by our correction g_{j-s}, whose valuation is $3 - 2(j-s)$.

For $r \geq j$, the standard coordinates at Z_r vanish; positivity after clearing the denominator excludes the corresponding full-spherical canonical summands. At $r = j - 1$ the coordinate starts in degree one, so its bar-symmetric spherical multiplicity is zero. Now descend through $r \leq j - 2$. At Z_r all one-incidence terms in Y_j, including their multiplicities, have valuation at least three (step 2 and A Lemma 4H.1). Every family-F coefficient there is proper, hence has positive valuation, and its multiplicity is constant. Full-spherical indices of rank less than r have zero coordinate. Thus after subtracting the already known higher full-spherical terms, the nonpositive powers at Z_r determine its bar-symmetric multiplicity completely. By step 3 and the preceding Gaussian comparison these powers agree with our smaller-parameter computation (Lemmas 4J.2, 4K.1). It follows that the multiplicity at Z_r in Y_j is g_{j-r} for $0 \leq r \leq j-2$, and zero for $r \geq j-1$. At $r = 0$, after all these subtractions the coefficient of v is also the one in our smaller calculation, namely $\binom{j+1}{2}$. For $j = 1$, the raw coordinate begins with v, and there is no spherical subtraction; this gives the same formula directly.

The only remaining contributors in degree one at Z_0 are the family-F terms of step 2. We have proved $\sum_{i=1}^{j} \mu_{i;m} = \binom{j+1}{2}$ while *holding* m *fixed*. Taking consecutive differences gives the individual formula.

Equivalently, before reading the distinguished coefficient, $X_i T = Y_i - Y_{i-1}$ contains $C_{A_{i;m}}$ once and none of the other family-F vectors. Its full-spherical coefficient at Z_r is $g_{i-r} - g_{i-1-r}$, with $g_a = 0$ for $a \leq 1$. The residual one-incidence terms can persist as operators. Their origin coordinates, including multiplicities, start in degree at least three by the same cleared-positivity bound. We thus identify one canonical summand in the parabolic product for each entry of our row.

5 The first-entry polynomial and rank reduction

Put $b_{j;m} = u_{j;m}^{-1}$ and $h_x = [M_e]\mathbf{C}_x$. In the word convention of the parabolic paper, the negative, zero and positive shifts in our sorted window give the letters $\mathsf{C}, \mathsf{B}, \mathsf{A}$, respectively. Thus the word of $b_{j;m}$ is

$$\mathsf{C}^{j-1}\mathsf{B}^{m-j}\mathsf{ABC}\mathsf{B}^{m-j}\mathsf{A}^{j-1}.$$

The two reflection chains in that paper keep m letters of each wound type throughout. The off-diagonal endpoints have $j < m$ of each.

The first entry admits a finite Hecke calculation

Deduction. For every $m \geq 2$,

$$P_{w_0,C_{1;m}}(q) = 1 + q + \cdots + q^{m-1}, \qquad \mu_{1;m} = 1.$$

Indeed, the window of s_0 has 0 at position 1 and $N+1$ at position N. Moving them to positions $m, m+2$ gives

$$b = b_{1;m} = s_0(s_1 \cdots s_{m-1})(s_{N-1} \cdots s_{m+2}) \in W_J.$$

Use the same $V = I \cap J$ and $L = V \setminus \{s_1, s_{N-1}\}$ as in our gallery. Thus $W_V = S_m \times S_m$. Write $[a] = (v^a - v^{-a})/(v - v^{-1})$. In the spherical right module consider the bar-invariant vector $G = M_e C_{s_0} C_V/\pi_L$, where $C_V = C_{w_V}$. The right W_V-orbit of the left coset indexed by s_0 has minimal index s_0 and maximal index b. Its elements move the one A within the first block of m positions and the one C within the last block, keeping the middle B fixed. The stabilizer is W_L. Expanding the parabolic symmetrizer into minimal coset representatives gives

$$G = \sum_{x \in \mathcal{O}_V(s_0)} v^{\ell(b)-\ell(x)} M_x + v[m]^2 M_e.$$

Here $M_e C_{s_0} = M_{s_0} + vM_e$, and $\pi_V/\pi_L = [m]^2$. For the orbit term, each shortest shuffle moves the two marked letters away from their initial positions by length-increasing steps. Thus it acts by $M_{s_0} H_z = M_{s_0 z}$, with exponent $\ell(w_V) - \ell(w_L) - \ell(z) = \ell(b) - \ell(s_0 z)$. This is also the orbit formula underlying B §3.3. Now $v[m]^2 - [m][m-1] = v^m[m]$. The vector

$$\mathbf{C}_b = G - [m][m-1]M_e$$

is characterized by bar invariance, leading standard M_b with coefficient one, and strictly positive powers of v at all other standards. This proves the displayed KL formula since $h_b = v^m[m]$ and $\ell(b) = 2m - 1$.

This finite calculation gives the entire first-column KL polynomial. For example, at $m = 3$ it gives $1 + q + q^2$. Section 4 determines the leading coefficient of every column.

The first seed in the uncorrected parabolic family

The same entry can be read from the parabolic product in the ambient period $2m + 1$. With T_1, T_2 as in $(*)$ and our strong one-line canonical $\mathcal{E}_{1,m}$ refined to V, one has

$$\mathcal{BS}_1 T_1 T_2 = \mathbf{C}_{b_{1;m}} + [m-1]\mathcal{E}_{1,m}.$$

Here $\mathcal{BS}_1 = D(J_1)$ has coefficient v^{2m-1} at e and v^{c-3} at the rank-one words with A in position one and C in position $3 \leq c \leq N$ (B Lemma 3.19). For the three output orbit maxima H, U, F, the A is in position m and the C in positions $1, m+1, m+2$, respectively. The four standard coordinates are

	H	U	F	e
$\mathcal{BS}_1 T_1 T_2$	$[m-1]$	$v^{m-1}[m-1]$	$v^{m-1}[m]$	$v^{2m-1}[m]^2$
$\mathcal{E}_{1,m}$	1	v^{m-1}	v^m	$v^{2m}[m]$.

For the first row, T_1 sends the terms $3 \leq c \leq m+1$ to their orbit sum with top coefficient $\sum_{c=3}^{m+1} v^{c-3+c-m-1} = [m-1]$, and multiplies the others by $[m]$. For H, the second averaging contributes the sum $\sum_{c=2}^{m} v^{c-2+c-m} = [m-1]$ divided by $[m-1]$; for U, F it moves a single A to the end of the first block. The second row follows from A Lemmas 4F.2 and 4D.1. These are all occupied orbits. Subtract the rows with the stated factor, using $v^{m-1}[m] - v^m[m-1] = 1$.

The rank-reduction identity

Let Θ_k have word $\mathsf{C}^k\mathsf{B}^{N-2k}\mathsf{A}^k$. It is the finite-orbit maximum of type k (B Lemma 3.11), hence represents our full-spherical standard Z_k after output refinement. The coefficient at its longest fine representative has shift zero (A Lemma 4D.1). For $0 \le k < j \le m$ we obtain the all-degree identity

$$[\Delta_{Z_k}]C_{A_{j;m}} = [M_{\Theta_k}]\mathbf{C}_{b_{j;m}} = h^{(m-k)}_{b_{j-k;m-k}},$$

where the superscript gives the smaller rank parameter. For $k > 0$, both ordinary endpoints $w_I\Theta_k, w_I b_{j;m}$ take the values $2N, 1-N$ at positions $1, N$, respectively. Every fundamental-window entry lies between those bounds. Deleting the common first and last letters removes one C and one A: the relabelling in B Definition 4.17 gives $w'_I\Theta_{k-1}, w'_I b_{j-1;m-1}$, with period $N-2$. To check Bruhat comparability use the rank functions $\#\{i \le a : w(i) \ge b\}$. At each cut the deleted periodic strands contribute equally to the two functions. On the remaining strands their inequalities are exactly the smaller-period inequalities. Iterating reaches the comparable pair $w'_I, w'_I b_{j-k;m-k}$. B Proposition 4.18 now applies at each deletion and preserves the normalized coefficient. For $j \le k$ the original coordinate vanishes since $\ell(\Theta_k) = 2k(N-k) \ge 2j(N-j) > D_{j;m}$.

On the diagonal, the coefficient formula in the parabolic paper gives $[v][\Delta_{Z_k}]C_{A_{m;m}} = m-k$ for $0 \le k < m$. The final case reduces to s_0 in period three. The independent KL recursion checked the full polynomial identity for every k, j in ranks $m = 2, 3, 4$. For example, at $m = 4$ the rows of linear coefficients, with columns $j = 1, 2, 3, 4$, are $(1, 2, 3, 4), (0, 1, 2, 3), (0, 0, 1, 2), (0, 0, 0, 1)$. Together with our one-line degree bound at each Z_k (Lemma 4H.1), these tests at full-spherical standards recover the row sums in parameters $m-k$. To distinguish entries within a fixed row, Section 4 computes its successive partial sums.

The length gap in direct rank reduction for $2 \le j < m$

At the ordinary pair $w_I, w_I b_{j;m}$ the only common window position is the central B at $m+1$. The other two B-blocks have $b(i) = i+1$ and $i-1$, respectively. Deleting the common position and image leaves type j in period $N-1$. In B Lemma 3.9, $\ell = 2j(N-j) - \mathrm{inv}$ with $\mathsf{C} < \mathsf{B} < \mathsf{A}$. Deletion drops two inversions, one with the preceding A, one with the following C. The ordinary length gap therefore drops by $2j-2 > 0$, preventing Henderson cancellation at this strand. Across fixed j the full gap difference is $D_{j;m} - D_{j;j} = (4j-2)(m-j)$.

The seed correction also depends on the ambient parameter. B Proposition 3.21 gives lowest exponent $2-(N-2j) = 1-2(m-j) < 0$ for $\mathcal{BS}_j$ at the orbit maximum $z(j,1,1)$. The terms $\mathcal{BS}_{j-1}$ and $\mathcal{CBS}_s$ for $s \le j-1$ vanish on this type-j orbit. Replacing m by j in the correction while retaining period $2m+1$ therefore leaves a negative power at this orbit, so it does not give the canonical seed.

6 What follows for coefficients and cohomology

Write $k = \overline{\mathbb{F}}_p$, $N = 2m+1$, and $G = \mathrm{SL}_N(k)$. For degree one, both papers assume Lusztig's character formula in the given rank and $p \geq 5(N-1)$ for finite group comparison. These assumptions hold for every sufficiently large p at fixed m.

Setting and source	Consequence
A, Theorem 4L.1 and Corollary 6C.2	$\sum_{j=1}^{m} \mu_{j;m} = m(m+1)/2$. Each coefficient equals the rational H^1-dimension of $L(\lambda_{j;m}(p))$ and the finite H^1-dimension for its restriction to $\mathrm{SL}_N(\mathbb{F}_p)$ and its descendant $M_{j;m}(p)$ on $\mathrm{PSL}_N(\mathbb{F}_p)$, over k. The row sum gives $\max_j \mu_{j;m} \geq \lceil (m+1)/2 \rceil$.
B, Theorems A and B	$\mu_{m;m} = m$, and $\dim_{\mathbb{F}_p} H^1(\mathrm{PSL}_N(\mathbb{F}_p), V_{m,p}) = m$. All $V_{m,p}$ are faithful and absolutely irreducible.
The two KL theorems together	$\sum_{j<m} \mu_{j;m} = m(m-1)/2$ and $\max_j \mu_{j;m} \geq m$.
Partial-row calculation, §4	$\mu_{j;m} = j$ throughout, with maximum m at $j = m$. For each sufficiently large p, our m modules realize every first cohomology dimension $1, \ldots, m$ for this one finite group.
B, Theorem B, (2.17), Corollary 2.20	For $0 \leq s < m$, the normalized polynomial for y_s starts with mv^{s+1}. For large p, $\dim_k H^{s+1}(G, L(y_s \cdot_p 0)) = m$, and its rational cohomology vanishes in all lower degrees. The identified diagonal is $s = 0$.
B, Theorem C, Lemma 2.18	Fix $1 \leq n \leq m$. For sufficiently large p, then sufficiently large a, a faithful absolutely irreducible k-module for $\mathrm{PSL}_N(\mathbb{F}_{p^a})$ has $\dim_k H^n \geq m$. The proof uses injective restriction.
B, Theorem D, (2.18)	For a primitive complex root of odd order $\ell > N$, $\dim_{\mathbb{C}} H^{s+1}(U_\zeta, L_\zeta(y_s \cdot_\ell 0)) = m$, with vanishing in lower degrees. This is Ext in finite-dimensional integrable type-1 modules of Lusztig's divided-power quantum group.
A with B's (2.18)	The same standard quantum KL formula transports our row sum to quantum H^1 for the m modules $L_\zeta(u_{j;m}^{-1} \cdot_\ell 0)$.

The coefficient fields. On the diagonal the highest weights agree. The split group gives an $\mathbb{F}_p$-form of the simple highest-weight module. Its root-lattice weight makes the finite center act trivially. That center has prime-to-p order, so inflation across it is an isomorphism in every degree. Thus, as modules for the *same* finite group,

$$M_{m;m}(p) \cong V_{m,p} \otimes_{\mathbb{F}_p} k.$$

Extending the finite group bar complex shows that the scalar extension preserves the cohomology dimension. The same descent applies to our other restricted highest weights. See A Lemmas 6A.2, 6B.1 and 6C.1; B Proposition 2.15 and the proof of Theorem A.

Finite-group comparison in higher degrees. For the weights in the parabolic paper, the standard large-characteristic comparison gives exact finite-group cohomology dimensions. Fix $1 \leq n \leq m$, set $\lambda = y_{n-1} \cdot_p 0$, and assume the character-formula hypotheses in that paper together with $p \geq (4n+1)(N-1)$. B Lemma 2.17 makes λ nonzero, p-restricted and a root-lattice weight in the affine dot orbit of zero. It is therefore p^a-restricted for every $a \geq 1$. Parshall–Scott–Stewart, *Shifted generic cohomology*, Theorem 5.12 and its proof on pp. 1780–1781, give

$$H^d(\mathrm{SL}_N(\mathbb{F}_{p^a}), L(\lambda)) \cong H^d(G, L(\lambda)) \qquad (1 \leq d \leq n,\ a \geq 1).$$

Here one takes $\nu = 0$ in the Bendel–Nakano–Pillen Theorem 7.5 comparison displayed in that proof. Combining this with Theorem B and (2.17) of the parabolic paper, and inflating across the prime-to-p center, gives a faithful absolutely irreducible $k\mathrm{PSL}_N(\mathbb{F}_{p^a})$-module V_a with

$$\dim_k H^n(\mathrm{PSL}_N(\mathbb{F}_{p^a}), V_a) = m, \qquad H^d(\mathrm{PSL}_N(\mathbb{F}_{p^a}), V_a) = 0 \quad (0 \leq d < n).$$

Degree zero vanishes by nontrivial irreducibility. In particular, equality and lower-degree vanishing hold for $a = 1$. This follows from the rational cohomology calculation in the parabolic paper and the cited comparison theorem. The required lower bound on the characteristic depends on the rank and the chosen degree. doi:10.1112/S0010437X13007331.

7 Exact calculations in the spherical module

The accompanying Python program implements two calculations over $\mathbb{Z}[v, v^{-1}]$. In the first, represent a standard vector by the levels $a_i = \lfloor (x(i) - 1)/N \rfloor$ of the left-minimal window. To apply C_P/π_T, where T is the previous intersection, cut the circle at a missing reflection and add one to the levels that pass the cut. In each position block sort the lifted levels in increasing order. Let L be the stabilizer of this minimum in W_P, $d(x)$ the within-block inversion number and $D = \ell(w_P) - \ell(w_L)$. The new coefficient at y in its orbit O is

$$v^{D-d(y)} \frac{\pi_L}{\pi_T} \sum_{x \in O} v^{-d(x)} [M_x] F.$$

Indeed $M_x C_P = v^{-d(x)} M_{\min O} C_P$ and splitting W_P into minimal left W_L-coset representatives evaluates the latter expression. All Laurent divisions in the program check for zero remainder.

The second calculation uses only right simple reflections. For a right descent s of x, recursively calculate $\mathbf{C}_{xs} C_s$. At each strict lower z subtract the constant term of its standard coordinate times $\mathbf{C}_z$. The resulting monic bar-invariant vector satisfies the canonical triangularity condition. This recursion independently computes the seed in the parabolic paper, every endpoint, and the canonical transition matrix used to decompose the products. The numerical inputs are the parabolic sets, the chain words and the endpoint windows.

Let $L_m = C_{s_2} \cdots C_{s_m} C_{s_1} \cdots C_{s_{m-1}}$ and let D_m denote the two normalized parabolic averages. Each cell below gives the degree-one identity coefficient followed by the number of distinct canonical indices with nonzero graded multiplicity.

m	$\mathcal{R}_m$	$\mathcal{BS}_m L_m$	$\mathbf{C}_{X_0} L_m$	$\mathbf{C}_{X_0} D_m$	$\mathbf{C}_{\beta_m}$
2	4, 5	4, 5	3, 3	3, 3	2, 1
3	11, 11	18, 23	12, 13	7, 6	3, 1
4	25, 17	81, 70	50, 38	14, 9	4, 1

In each rank all five columns have identical standard support, respectively 38, 368, 3098 indices, and coefficient one at M_{β_m}. Every coordinate of $\mathcal{BS}_m D_m$ agrees with $\mathcal{R}_m$. Of the family-F summands, the first two columns contain the entire row with multiplicity one, and the diagonal is the unique family-F term in the last three. The independent KL recursion gives rows $(1, 2), (1, 2, 3), (1, 2, 3, 4)$. For instance

$$[M_e]\mathbf{C}_{\beta_2} = 2v + 4v^3 + 5v^5 + 4v^7 + v^9, \qquad [M_e]\mathbf{C}_{X_0} = v + v^3 + 2v^5 + v^7 \quad (m = 2).$$

Executing `python3 compare.py -full 2 3 4` in the directory containing `compare.py` reproduces the tables and full identity polynomials in `results.txt` without external packages. A second script, `python3 partial.py 2 3 4 5`, checks the mixed full-spherical coordinates and the discarded degrees directly. In rank five its successive corrected degree-one coefficients are $1, 3, 6, 10, 15$; the independent KL recursion gives $(1, 2, 3, 4, 5)$. For selected entries at larger values of m, a separate run gives $\mu_{2;6} = \mu_{2;7} = \mu_{2;8} = 2$, $\mu_{3;6} = \mu_{3;7} = 3$ and $\mu_{4;6} = 4$. The last expansion has 120 862 standard terms. The script `check_off_diagonal.py` and its output accompany the calculation.

Comparison after boundary projection. At $m = 2$, $L_m = D_m$. For $m \geq 3$, applying the projector C_V/π_V to $\mathcal{BS}_m L_m$ gives a vector linearly independent of $\mathcal{R}_m$ over $\mathbb{Q}(v)$. It suffices to compare two coordinates at $v = 1$. Write $t = 2^{m-1}/m$. The ratio at M_e is t^2 since each simple factor multiplies this coordinate by 2, whereas each divided parabolic factor multiplies it by m. The ratio at M_{β_m} is t, as follows.

In type m the first affine product has just the word $A^m B C^m$. The next K_m-average fixes the initial A and puts the B in any one of positions $2, \ldots, m+1$. The E-average then gives coefficient one to every type-m word with prefix AB (the stabilizer of the remaining letters has size $(m-1)!m!$, exactly the overlap denominator). The last m letters stay fixed during L_m. In the V-orbit of β_m they contain one C and $m-1$ letters A, so only the initial prefix ABC^{m-1} of length $m+1$ can contribute: the leftmost letter is A and the second letter cannot be C. All first-chain swaps must be chosen to put B at position $m+1$. Each of the 2^{m-1} choices in the second chain then contributes. There are m possible suffixes and m^2 words in the target orbit. The normalized V-average gives $m2^{m-1}/m^2 = t$ at M_{β_m}, where $\mathcal{R}_m$ has coefficient one. The denominators and the coordinate used to compare scalars are nonzero at 1. Since $t > 1$, the two ratios differ.

Partial galleries. For $T_j = \mathcal{BS}_j D_m$ with $1 \leq j \leq m \leq 4$, exact elimination gives family-F multiplicity one exactly at indices $1, \ldots, j$. A truncated gallery can also contain canonical indices outside the six families for $\mathcal{R}_m$. At $(m, j) = (3, 2)$ its top word is $CCABBBA$, and the family-F word $CBABCBA$ occurs in a lower summand. Section 4 singles out the positive summand supported on the relevant seed indices in every rank.

Affine-window implementation

The directory `comparison/independent_checks/` contains an exact affine-window implementation separate from the level-word implementation above. Its inputs are the defining Hecke relations, parabolic subsets and endpoint windows. The recursion constructs canonical vectors directly from these inputs, giving an independent check on the row formula and the canonical expansions.

The engine's validation checks quadratic, braid and commuting relations, inversion, and parabolic-quotient enumeration. The normalized partial-sum calculation at $m = 2, 3, 4, 5$ checks every standard coordinate of $\mathcal{BS}_m T_1 T_2 = \mathcal{R}_m$, every family-F multiplicity in every Y_j, every spherical correction g_{j-r}, and the degree bounds on the remaining terms. The corrected partial rows are $1, 3$, $1, 3, 6$, $1, 3, 6, 10$, and $1, 3, 6, 10, 15$. The finite-tail calculation checks the canonical identities in Section 2.2 through $m = 5$, including all lower correction terms.

The scripts and their saved outputs make these exact Python calculations reproducible. They check finite instances of the uniform arguments given in the note and are separate from our Lean development.

8 A question about growth

The parabolic endpoint β_m is the last member of our row. For each sufficiently large prime p, the modules in the row have first cohomology dimensions $1, \ldots, m$ for $\mathrm{PSL}_{2m+1}(\mathbb{F}_p)$, of Lie rank $2m$. Can one find a family for which the first cohomology dimension grows faster than linearly with the rank? We would be glad to explore this question together.